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A Spatially Weighted Neural Network Based Water Quality Assessment Method for Large-Scale Coastal Areas

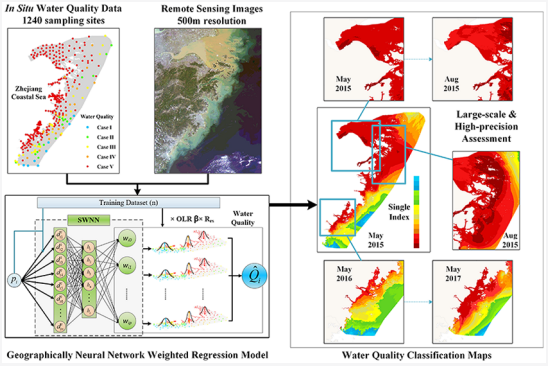
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ABSTRACT: The accurate assessment of large-scale and complex coastal waters is a grand challenge due to the spatial nonstationarity and complex nonlinearity involved in integrating remote sensing and *in situ* data. We developed a water quality assessment method based on a newly proposed geographically neural network weighted regression (GNNWR) model to address that challenge and obtained a highly accurate and realistic water quality distribution on the basis of the comprehensive index of Chinese Water Quality Classification Standards. Using geostationary ocean color imager (GOCI) data and observations from 1240 water quality sampling sites, we conducted experiments for a typical large-scale coastal area of the Zhejiang Coastal Sea (ZCS), People's Republic of China. The GNNWR model achieved higher prediction performance (average R^2 = 84%) in comparison to the widely used models, and the obtained water quality classification (WQC) maps in May of 2015−2017 and August 2015 can depict intuitively reasonable spatiotemporal patterns of water quality in the ZCS. Furthermore, an analysis of WQC maps successfully illustrated how terrestrial discharges, anthropogenic activities, and seasonal changes influenced the coastal environment in the ZCS. Finally, we identified essential regions and provided targeted regulatory interventions for them to facilitate the management and restoration of large-scale and complex coastal environments.



KEYWORDS: *Water quality assessment, Coastal water quality, Large-scale areas, Remote sensing, Geographically neural network weighted regression, Spatial nonstationarity*

1. INTRODUCTION

Healthy coastal water environments have many benefits for the human inhabitants of these regions; they improve health and offer opportunities for recreation, fishing, and industry.[1−3] However, the degradation of coastal water quality has caused serious damage to ecosystem functioning, and hypoxia zones have spread exponentially in coastal waters since the 1960s.[4] These large-scale water quality changes in coastal marine environments have attracted considerable attention from policy makers and scientists. The State Oceanic Administration of China (SOA) established a number of water quality assessment systems in 1997, including monitoring plans and water quality standards for coastal marine environments.[5,6] The European Water Framework Directive (WFD), which was adopted in 2000, made aquatic ecology the basis of management decisions for coastal bays, lagoons, and archipelago areas.[7,8] Studies have shown that unfavorable changes in coastal water quality pose threats to species and public health, and the degradation of water quality may become widespread without effective environmental management strategies.[9−11] Consequently, large-scale water quality assessment and targeted regulatory interventions are increasingly necessary to preserve coastal ecosystems.

Traditional monitoring methods, including *in situ* sampling and drifting instruments, are limited in spatial coverage. Because water quality varies across space, making use of large-scale monitoring data to investigate the detailed spatial distribution of coastal water quality is crucial for making accurate, pinpointed regulatory interventions.[7,12] Remote sensing monitoring is a powerful tool for the observation of large-scale waters and is widely used to evaluate environmental processes.[13,14] Recent studies have indicated that estimating the ecological characteristics of the waters using satellite data is justified, and the integration of the remote sensing and *in situ* data is a feasible approach to assess water quality on a large spatial scale.[15−17]

Previous research has mainly focused on the retrieval of one or more specific water quality parameters by integrating remote sensing reflectance (R_{rs}) and *in situ* data.[18,19] However,

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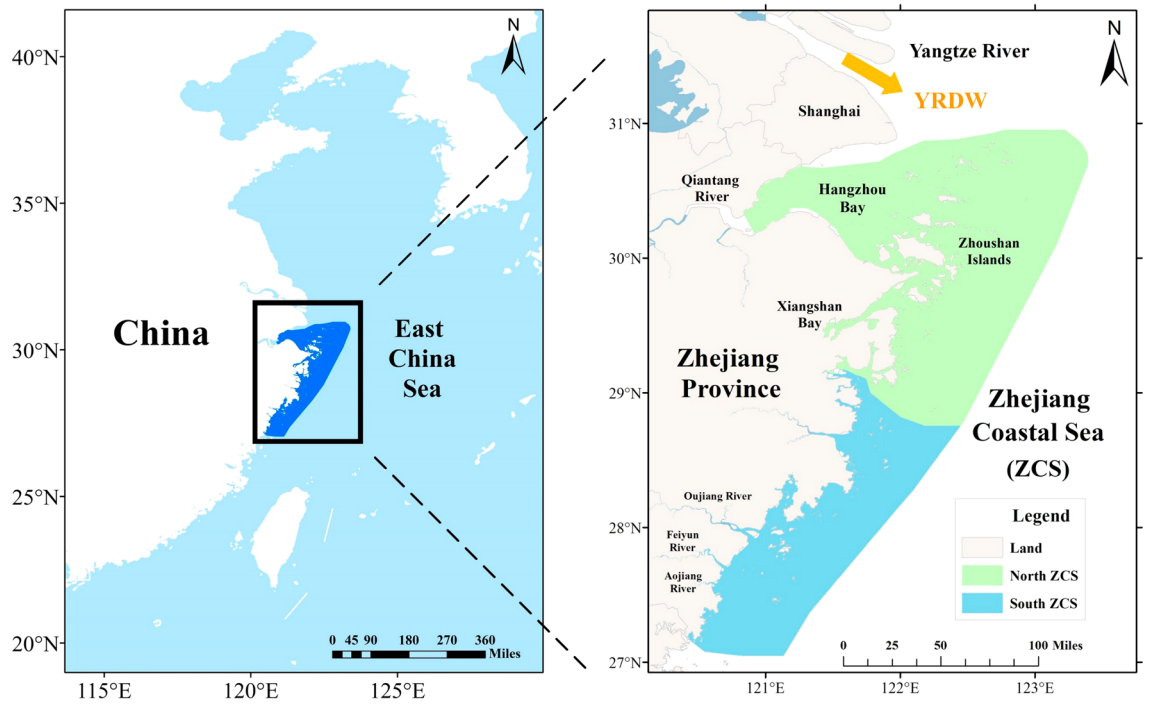


Figure 1. Location of the study area. The Zhejiang Coastal Sea (ZCS) is located in the East China Sea and can be divided into the north ZCS and south ZCS. Hangzhou Bay, Xiangshan Bay, and the Zhoushan Islands are rich in marine resources. The orange arrow indicates the rough direction of the Yangtze River Diluted Water.

using a single, comprehensive assessment index to manifest water quality can make it simpler for environmental managers and the general public to understand and can thus directly promote water management practices.[20] For example, band-based empirical models may be able to accurately retrieve specific constitutes in coastal waters, but there are no such models that aim to obtain a comprehensive assessment index.

Coastal waters are influenced by a number of environmental factors, such as terrestrial discharges and anthropogenic pollution, that may be spread extensively by rivers flows, tides, and ocean currents. Therefore, the relationships between water quality and R_{rs} might vary continually over space (i.e., spatial nonstationarity) and exhibit complex nonlinearity, especially in large-scale waters and complex environments.[21−24] Previous studies have indicated that traditional linear models and empirical models cannot handle these problems.[10,25] Although spatial regression models (e.g., geographically weighted regression, GWR) can estimate the spatially nonstationary relationship between water quality and R_{rs}, they can only depict simple nonlinear structures.[26,27] In addition, previous algorithms were mainly developed for inland waters, open seas, rivers, or semienclosed bays. The modeling of large-scale water quality is, however, a complex nonlinear problem due to the wide range of land−ocean interactions.[25,27] Artificial neural network (ANN) algorithms can address complex nonlinear problems by exploiting their greater calculability.[1,28] Nazeer et al. used back-propagation algorithms to estimate Chl-a and suspended solids on the coastal sea of Hong Kong and achieved R^2 values of 0.59−0.77.[29] Ioannou et al. used a multilayer perceptron neural network (MLPNN) to retrieve the inherent optical properties (IOP) from R_{rs} at available MODIS wavelengths and achieved R^2 values of 0.78−0.9 with a significant amount of training.[30] However, the spatial predictions of ANN-based models may exhibit unreasonable patterns and their established relationships may be spatially uninterpretable due to the lack of consideration of spatial effects.[31] Further, ANN models often need to add supplemental water quality data to improve the prediction accuracy.[28,32]

Therefore, to date, no study has directly provided a water quality assessment index in large-scale and complex coastal waters through effective remote sensing modeling. There is an urgent need for a new method to deal with both spatial nonstationarity and complex nonlinearity in coastal water quality modeling. In view of this, this study aims to map a water quality assessment index in large-scale coastal waters using remote sensing retrievals through a newly developed geographically neural network weighted regression (GNNWR) model.[27] The GNNWR, which integrates ordinary linear regression (OLR) and a spatially weighted neural network (SWNN), can simultaneously address spatial nonstationarity and complex nonlinearity and thus has the potential to enhance the accuracy and reasonableness of water quality assessment.[27]

The Zhejiang Coastal Sea (ZCS), as shown in Figure 1, is a typical coastal area located in the East China Sea. It has been suffered from large-scale eutrophication since 1980s.[33−35] In this study, we took the ZCS as the study region for three reasons.

(1) The ZCS is a complex large-scale coastal water, with a sea area of more than 44400 km^2 and a coastline of 6500 km. Previous studies have indicated that the water quality parameters in the ZCS show spatial nonstationarity, and nonlinear modeling is required for this complex environment.[21,26,36]

(2) In China, the SOA classified the water quality of *in situ* sites into five levels under the Chinese Water Quality Classification Standards GB3097-1997 (CWQCS),

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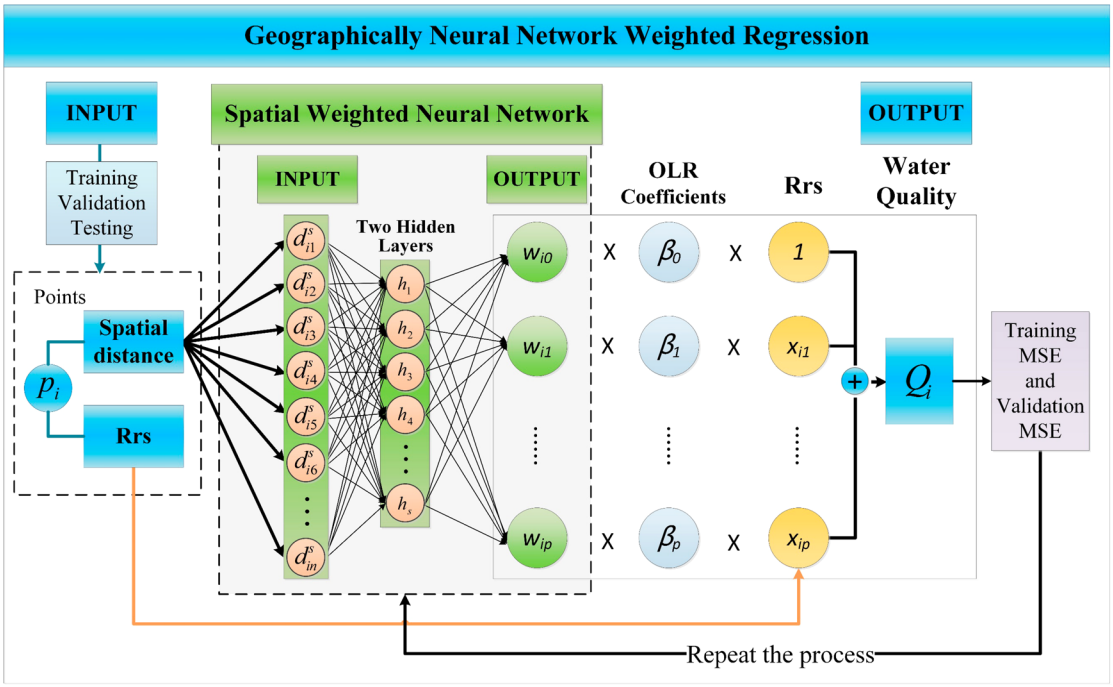


Figure 2. Construction of the GNNWR model and SWNN.

which can serve as a simple, comprehensive index to manifest water quality and provide an intuitive characterization of large-scale waters.[5,37]

(3) More than 58 ports, 8 rivers, and 500 marine functional zones are located in the ZCS, which make it an intensive area for aquaculture, tourism, shipping, and industry.[38,39] Exploring this human-impacted ecosystem is meaningful for the sustainable development of the marine economy.

In this study, through the integrated application of the GNNWR model and CWQCS index, we aimed to obtain water quality classification (WQC) maps for the ZCS in May of 2015−2017 and in August 2015. Geostationary ocean color imager (GOCI) data and 1240 *in situ* sampling data were used in the experiments. We conduct the following analyses to demonstrate the proposed method as an effective tool for water quality assessment and management practice in large-scale and complex coastal waters.

(1) To verify the precision of the model, we compare the GNNWR model with other models by performance indicators, including OLR, GWR, inverse distance weighting (IDW), and MLPNN models.

(2) To confirm that the large-scale water quality distributions in our WQC maps are reasonable, we compare and analyze the spatial patterns and gradient variations of water quality.

(3) To demonstrate the usefulness of our method, we analyze the WQC maps in detail in terms of the marine functional zones, nutrient discharges, river pollutants, and marine currents. Furthermore, we propose targeted water pollution regulatory interventions and other environmental suggestions based on our results.

2. MATERIALS AND METHODS

2.1. *In Situ* Water Quality Data. The Ocean Monitoring and Forecasting Center (OMFC) in Zhejiang province (Hangzhou, People's Republic of China) conducts quarterly *in situ* measurements in the ZCS using scientific vessels equipped with positioning systems. Because the ZCS has a high risk of algal blooms during spring, our experiments explored the water quality in May of 2015−2017. Water quality in the summer of 2015 (August), which has large riverine runoff, was also studied as a comparison. We used 307, 311, 314, and 308 sampling sites in May and August 2015, May 2016, and May 2017, respectively, as shown in Figure S1 in the Supporting Information. The parameters collected are chemical oxygen demand (COD), dissolved oxygen (DO), dissolved inorganic phosphorus (DIP), dissolved inorganic nitrogen (DIN), oil, and pH. According to the CWQCS, these parameters are merged into a single water quality index, i.e. water quality level, and the level of a site is determined by the lowest level of its water quality parameters. More detailed descriptions are given in Table S1 and S2.

2.2. Remote Sensing Data. The satellite data used in this research are the freely available GOCI data from the Korea Ocean Satellite Center (http://kosc.kiost.ac.kr), which provide images with a spatial resolution of 500 m at hourly intervals up to eight times a day. This data set provides observations for the whole of the ZCS and permits the monitoring of eight bands from visible to near-infrared.[40−42] We selected cloud-free data (level 1B) collected in the same month as the *in situ* data and preprocessed the images using the standard atmospheric correlation in the GOCI Data Processing System (GDPS).[43,44]

2.3. GNNWR Model. The GNNWR model is a spatial regression model in which the regression coefficient varies continually across space by integrating the OLR model and a SWNN.[27] The regression coefficient of the OLR model is the best unbiased estimate of all of the sample points:

$$Q_i = \beta_0 + \sum_{k=1}^{p} \beta_k x_{ik} + \varepsilon_i \quad i = 1, 2, \ldots, n \tag{1}$$

where the dependent variable and the kth independent variable are denoted as Q_i and x_{ik}, respectively. Accordingly, β_0 and β_k

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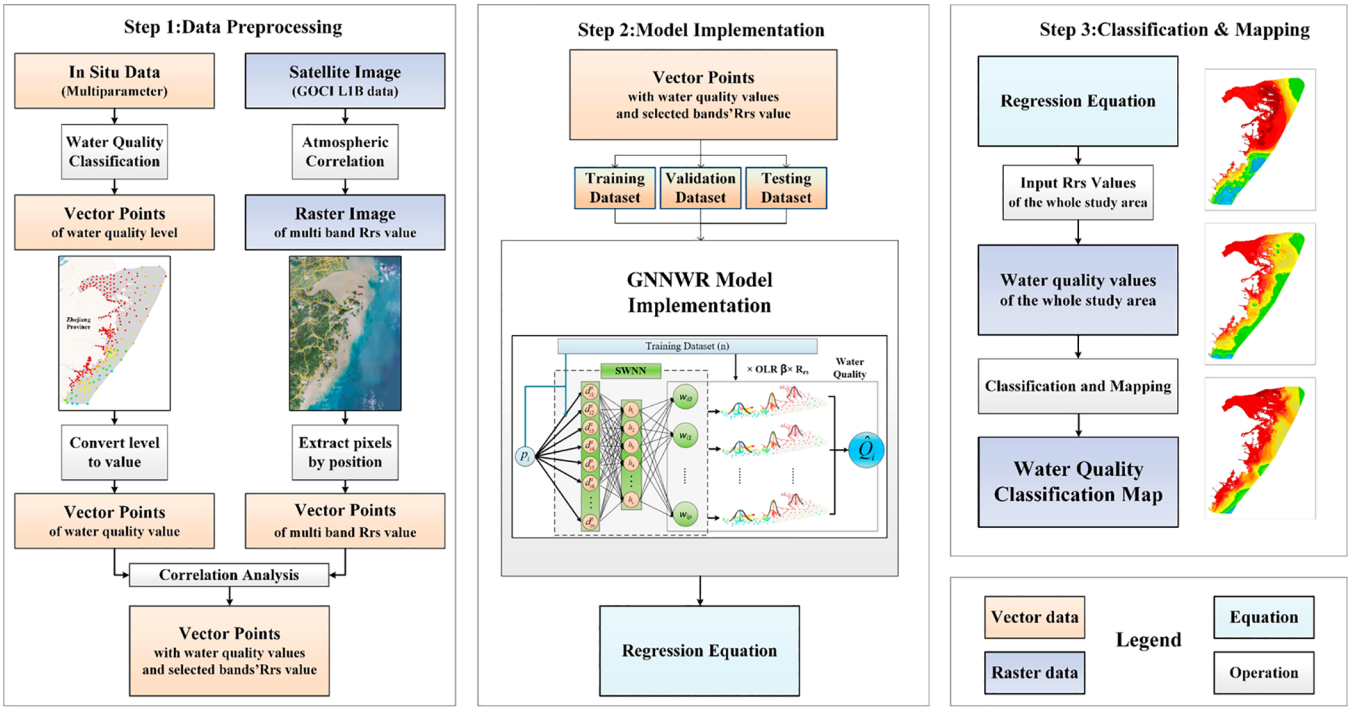


Figure 3. Water quality assessment method proposed in this paper, which adopts the CWQCS and uses the GNNWR model. This experiment took the *in situ* water quality level data and the satellite images of GOCI Level 1B data as the input and generated WQC maps of the Zhejiang Coastal Sea.

represent the constant coefficient and the kth regression coefficient of the corresponding independent variables, which reflect a global average relationship within the study area. Because the spatial nonstationarity can be regarded as spatial fluctuations of this average relationship, GNNWR defines the spatial nonstationary relationship as

$$Q_i = \sum_{k=0}^{p} \beta_k(u_i, v_i)x_{ik} + \varepsilon_i = w_0(u_i, v_i)\beta_0 + \sum_{k=1}^{p} w_k(u_i, v_i)\beta_k x_{ik} + \varepsilon_i \quad i = 1, 2, ..., n \tag{2}$$

where $w_0(u_i,v_i)$ and $w_k(u_i,v_i)$ represent the nonstationary weights of the OLR coefficients. To exactly calculate the fluctuating weights over space, GNNWR designates the SWNN as

$$\boldsymbol{W}(u_i, v_i) = \mathrm{SWNN}([d_{i1}, d_{i2}, ..., d_{in}]^T) \tag{3}$$

where $[d_{i1}, d_{i2}, ..., d_{in}]$ are the distances from the estimated location i to the training sample. As shown in Figure 2, the input layer of the SWNN comprises the spatial distances and its output layer comprises the spatial weights (i.e., $w(u_i,v_i)$).

In this study, the SWNN is a fully connected back-propagation network with two hidden layers and a stochastic gradient descent optimizer, which was implemented using the programming in TensorFlow 1.5.0. A more detailed description is provided in section 1.1 in the Supporting Information. The loss function of the SWNN is defined as the mean square error (MSE) of the estimated water quality value $\hat{Q}_i$ rather than that of the output weight:

$$\mathrm{loss} = \frac{\sum_{i=1}^{n}(Q_i - \hat{Q}_i)^2}{n} \tag{4}$$

2.4. Water Quality Assessment Method. As shown in Figure 3, the water quality assessment method involved three steps.

2.4.1. Step 1: Data Preprocessing. First, the *in situ* data were classified into five water quality levels from Case I to Case V according to the CWQCS. Then the five levels were converted into continuous values, as shown in the second column of Table 1. The continuous values were used as the dependent variable of the models, and therefore the outputs of the models are continuous water quality results that can be classified into more levels to show the detailed water quality variations. Next, the remote sensing image and the *in situ* points were transformed into the same projection and then all the R_{rs} values matching the spatial coordinates of the *in situ* points were extracted. Finally, correlation analysis was used to select the bands with higher correlations. In this experiment, bands 1, 3, 5, and 7 were finally selected, and the correlation analysis results are shown in Table S3.

2.4.2. Step 2: Model Implementation. The *in situ* points were randomly divided into training (70%), testing (15%), and validation (15%) data sets (see detailed information in Figure S2 and Table S4). The inputs of the GNNWR model are the points with continuous water quality values, R_{rs} values, and spatial coordinates. By implementing the models, we can obtain the regression relationship between water quality and R_{rs}.

2.4.3. Step 3: Classification and Mapping. The R_{rs} values in the whole study area were input into the regression equation to acquire the continuous water quality values. Then, these values were classified into 15 levels by ranges shown in the last column of Table 1, and the WQC maps were generated to show the detailed gradients of the water quality distribution in the 15 levels.

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Table 1. Water Quality Levels, Continuous Values, and Ranges[a]

water quality level	continuous value	new level	range
Case I	0.1	1st	≤0.067
		2nd	0.067−0.133
		3rd	0.133−0.200
Case II	0.3	4th	0.200−0.267
		5th	0.267−0.333
		6th	0.333−0.400
Case III	0.5	7th	0.400−0.467
		8th	0.467−0.533
		9th	0.533−0.600
Case IV	0.7	10th	0.600−0.667
		11th	0.667−0.733
		12th	0.733−0.800
Case V	0.9	13th	0.800−0.867
		14th	0.867−0.933
		15th	>0.933

[a]According to the Chinese Water Quality Classification Standards GB3097-1997, water quality is classified into five levels from Case I to Case V. This study converts these levels into continuous values from 0 to 1 (the second column) for model training. The outputted continuous values of the model are then reclassified into 15 levels (the third column) to show the detailed gradients of the water quality distribution. The ranges of the 15 levels are shown in the last column.

In addition, this study implemented the OLR, two GWR, MLPNN, and IDW models with the same data to compare the model accuracy (IDW only used *in situ* data). The environments of the model implementation are shown in Table S5. The specifications of the above models and evaluation indicators are provided in sections 1.1−1.5 in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Model Performance. Table S6 shows the fitting results of the OLR, GWR, MLPNN, and GNNWR models for the training data sets. A comparison of the results for each month reveals that all evaluation indicators of GNNWR are better than those of the other models, in that it has the highest R^2 value and lowest root-mean-square error. The indicators of GNNWR are significantly better than those of the other models in 2016 and 2017, illustrating its stable performance. Second, the p values of the GWR and GNNWR models are all 0.01, indicating that the regression relationship between water quality value and the R_{rs} value of the selected bands is significantly nonstationary in space. Moreover, GNNWR is superior to GWR in terms of the F_1 and the corrected Akaike information criterion value (eq S6 in the Supporting Information), suggesting that GNNWR has a better fitting effect for the spatial nonstationarity in comparison to GWR. In addition, the performance of the MLPNN model is unstable, as it shows higher accuracy in May 2016 and 2017 and lower accuracy in August 2015 in comparison with the two GWR models. These results demonstrate that the water quality in the ZCS has spatial nonstationarity and the GNNWR model has a better fitting ability in the regression modeling of coastal water quality.

Table 2 shows the spatial prediction results of the OLR, GWR, MLPNN, IDW, and GNNWR models for the testing data sets. GNNWR achieves the best indicators and IDW the second best, followed by the MLPNN and two GWR models, while the OLR results are still the worst. With May 2015 as an example, in comparison with the R^2 value (0.578) of the OLR, the GWR-AFG and GWR-AAB only increase the R^2 values to 0.631 and 0.663, the MLPNN and IDW increase the R^2 values to 0.701 and 0.743, and the GNNWR shows a significant improvement with the R^2 value increased to 0.826. In addition,

Table 2. Evaluation Results of the OLR, GWR, MLPNN, IDW, and GNNWR Models for the Testing Data Set

		R^2	RSS	MSE	RMSE	MAE	MAPE
May 2015	OLR	0.524	1.254	0.027	0.163	0.119	0.374
	GWR-AFG	0.696	0.800	0.017	0.130	0.070	0.272
	GWR-AAB	0.702	0.785	0.017	0.129	0.071	0.273
	MLPNN	0.747	0.667	0.014	0.119	0.079	0.246
	IDW	0.728	0.710	0.015	0.124	0.061	0.261
	GNNWR	0.832	0.444	0.009	0.097	0.051	0.051
Aug 2015	OLR	0.446	2.806	0.058	0.242	0.189	0.916
	GWR-AFG	0.810	0.964	0.020	0.142	0.095	0.367
	GWR-AAB	0.830	0.863	0.018	0.134	0.089	0.357
	MLPNN	0.596	2.049	0.043	0.207	0.157	0.677
	IDW	0.850	0.736	0.016	0.127	0.079	0.043
	GNNWR	0.885	0.584	0.012	0.110	0.083	0.344
May 2016	OLR	0.578	1.146	0.024	0.155	0.112	0.270
	GWR-AFG	0.663	0.917	0.019	0.138	0.101	0.258
	GWR-AAB	0.631	1.003	0.021	0.145	0.106	0.267
	MLPNN	0.701	0.814	0.017	0.130	0.087	0.232
	IDW	0.743	0.692	0.015	0.123	0.073	0.184
	GNNWR	0.826	0.472	0.010	0.099	0.074	0.174
May 2017	OLR	0.625	1.002	0.019	0.139	0.091	0.308
	GWR-AFG	0.658	0.914	0.018	0.133	0.082	0.277
	GWR-AAB	0.664	0.898	0.017	0.131	0.079	0.273
	MLPNN	0.688	0.835	0.016	0.127	0.075	0.234
	IDW	0.740	0.657	0.014	0.119	0.061	0.257
	GNNWR	0.816	0.492	0.009	0.097	0.053	0.181

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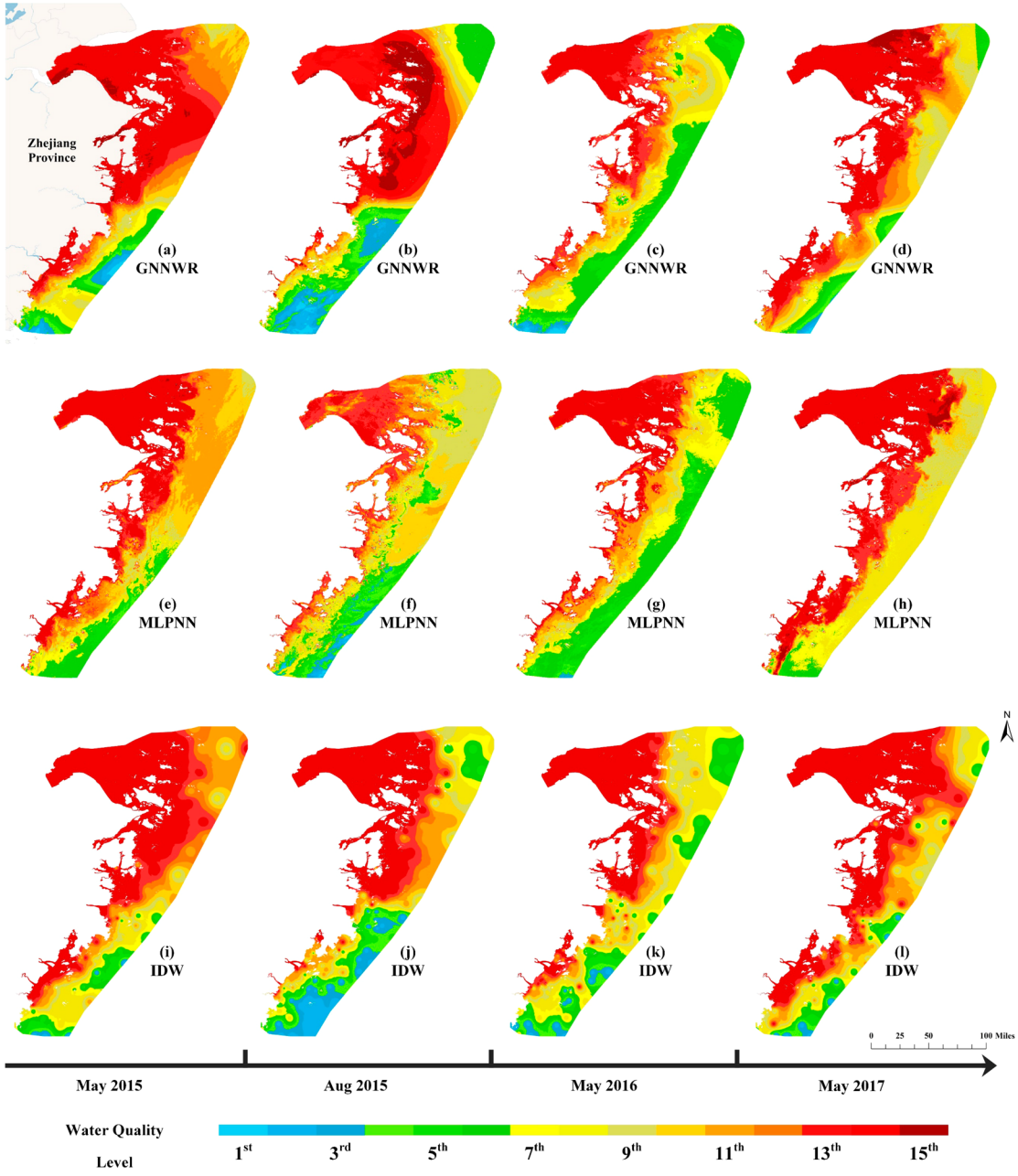


Figure 4. Water quality classification maps of the Zhejiang Coastal Sea (ZCS) in May and August 2015, May of 2016, and May 2017, respectively, acquired by the GNNWR, MLPNN, and IDW models. The water quality is classified into 15 levels and is represented by a comprehensive index based on Chinese Water Quality Classification Standards GB3097-1997. The colors from blue to red represent the water quality from the best to the worst. (a–d) give the water quality classification maps of ZCS obtained by the GNNWR model, (e–h) give the maps obtained by the MLPNN model, and (i–l) give the maps obtained by the IDW model.

in Aug 2015, the MLPNN has a very low R^2 value (0.596), which may be due to the lack of consideration of spatial effects, because the F_1 value of August 2015 is the lowest of the four months. Consequently, the GNNWR has achieved considerable accuracy and exhibits a superior spatial prediction, allowing us to address the complex relationship between water quality and R_{rs} in coastal waters.

3.2. Overall Water Quality Distributions in the ZCS.

Figure 4 shows the WQC maps obtained by GNNWR (Figure 4a–d), MLPNN (Figure 4e–h), and IDW (Figure 4i–l) in May 2015, August 2015, May 2016, and May 2017, which provide synoptic views of the water quality distribution in the ZCS. The WQC maps obtained by the OLR and GWR models are shown in Figure S3.

First, all of the maps intuitively show that the water quality distributions in ZCS vary greatly across space and time. In terms of the spatial distribution, the overall water quality in the north ZCS is worse than that in the south, which is consistent with the fact that the environment of the north ZCS is considerably affected.[38,45] According to the GNNWR maps, the area proportion of Case V water quality in the four months reached 54%, 51%, 28%, and 51%, respectively, which reflects that the water quality in the ZCS was quite poor in May 2015, August 2015, and May 2017.

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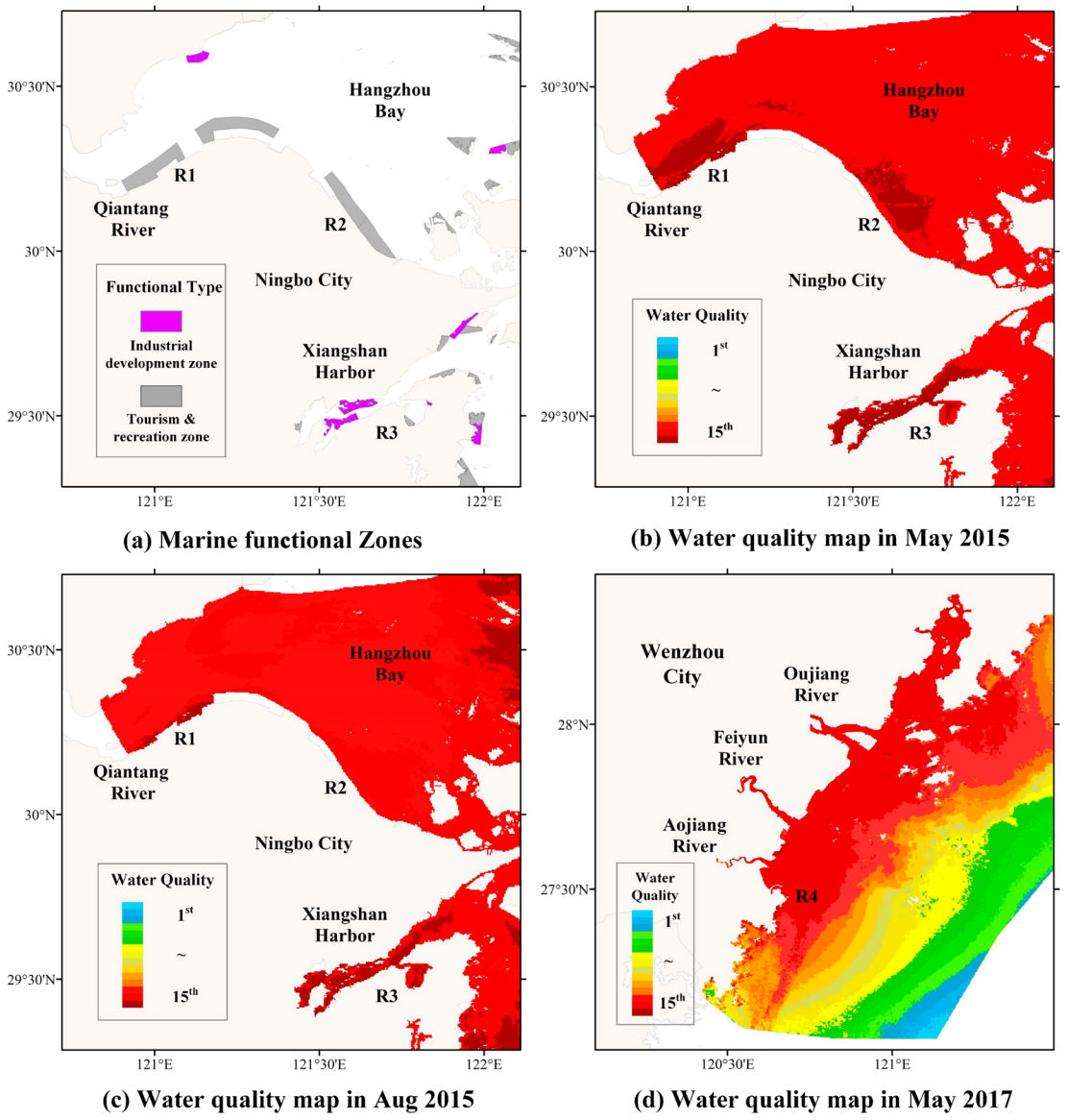


Figure 5. Enlarged maps used for detailed spatiotemporal analysis. (a) The marine functional zones in the Zhejiang Coastal Sea and (b, c) the water quality distributions in May and August 2015, respectively, enlarged on Hangzhou Bay and Xiangshan Bay. (d) The map in May 2017, enlarged on three key rivers.

Second, the IDW maps show accurate water quality in the regions near known points but the overall spatial distributions are too dependent on these points and show unsmooth variations across space. The GNNWR maps reflect the detailed gradients of the water environment, in which 15 levels of water quality are reasonably distributed from coast to open seas. In contrast, due to the lack of consideration of spatial effects, there are almost no instances of Case I water quality in the MLPNN maps and the water quality distributions are spatially uninterpretable. Therefore, the GNNWR maps appear to be more reliable and more consistent with the actual situation.

These preliminary results suggest that the GNNWR model has very good estimation accuracy on the basis of a comparison of performance indicators, and the GNNWR-generated WQC maps with a single water quality index offer a simple and reliable way for the general public and water managers to better understand the spatiotemporal patterns of water quality.[46] In section 1.6 in the Supporting Information, we analyzed the effect of individual parameters and find that DIN and DIP are dominant parameters affecting water quality in the ZCS. Therefore, the following section is more focused on the regions influenced by nitrogen and phosphorus and provides further evidence of the reliability and usefulness of the WQC maps.

3.3. Analyses of Essential Regions in the ZCS.

3.3.1. Water Quality Distribution in Two Inhabited Harbors. Many studies have shown that areas with high population densities and frequent anthropogenic activity are more likely to be influenced by nitrogen and phosphorus.[47,48] Therefore, we focused on Hangzhou Bay and Xiangshan Bay because of the densely inhabited cities around these two harbors.[49,50]

As shown in Figure 5a, marine functional zones, which are developed to serve the marine economy, are distributed in the two harbors. The inner regions of Hangzhou Bay (regions R1 and R2) in Figure 5b show level 15 water quality in May 2015, and these regions coincide with the two industrial development zones in Figure 5a. Thus, it can be inferred that the point-source pollution of the industrial emissions might be the

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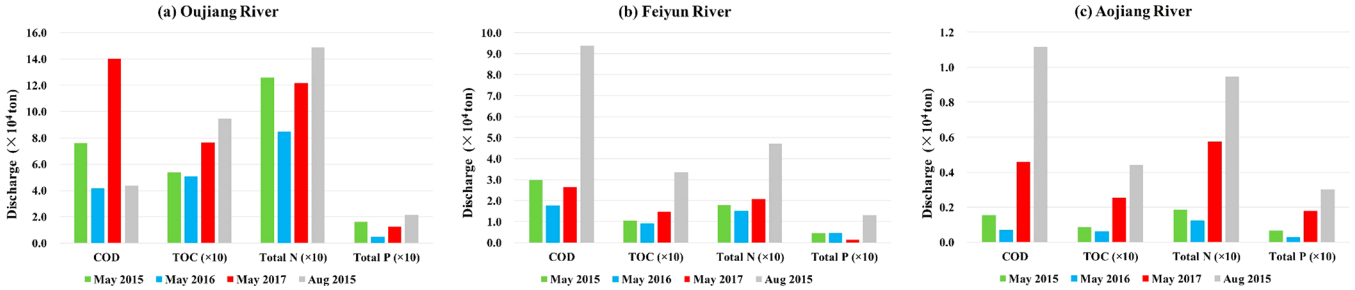


Figure 6. Estimated river pollutant data for the Oujiang River, Feiyun River, and Aojiang River in May of 2015−2017 and in August 2015, respectively. The factors are chemical oxygen demand (COD), total organic carbon (TOC), total nitrogen (total N), and total phosphorus (total P).

reason for the poor water quality in R1 and R2. Similarly, the inner Xiangshan Bay contains some level 15 regions (R3), which might be attributed to the non-point-source pollution of the tourism−recreation zones along the coast. In August 2015 (Figure 5c), R1 and R2 still show worse water quality than the adjacent areas, and R3 also presents level 15 regions, indicating that the poor quality in these regions was not underestimated by accident. In addition, as influenced by nitrogen and phosphorus, higher eutrophication indexes (section 1.7 and Table S7 in the Supporting Information) are found in the two months of 2015 and then decreased in 2016 and 2017, which is in agreement with the better water quality shown in regions R1−R3 (Figure 4c,d).[36,51]

The above analysis further confirms that the WQC maps generated by GNNWR are reliable and demonstrates that the spatiotemporal variations in the maps facilitate the characterization of the coastal water environment and the detection of abnormal water quality regions.

3.3.2. Water Quality Distribution in Three Adjacent Estuaries. In addition to marine development, anthropogenic pollutants that flow into rivers and accumulate in estuaries can influence coastal water environments.[52,53] As shown in Figure 6, we collected the river statistical data (nitrogen, phosphorus, COD, and TOC) of Oujiang River, Feiyun River, and Aojiang River to explore the effects of riverine discharges.

Figure 6 reveals that in May 2015, 2016, and 2017, most of the COD, TOC, and total N are the highest in 2017 (red columns), and all of the factors are the lowest in 2016 (blue columns). The estuaries in the south ZCS present a large area of Case V water in May 2017 (Figure 4d) and show the best distribution in May 2016 (Figure 4c), which indicates that the water quality in the three adjacent estuaries of the south ZCS is mainly controlled by the riverine discharges. In addition, it is worth noting that there is a jetlike water flow located in the south of the Aojiang River estuary (Figure 5d, R4), which has two possible explanations. First, previous studies reported that the Zhejiang-Fujian Coastal Current (ZFCC) and northeast monsoon carried waters from north to south along the ZCS in the spring; thus, the surface river pollutants tend to spread southward and generate a jetlike water flow.[32,54] Second, the high-concentration nutrients and organic matter in the riverine discharges in May 2017 might have resulted in severe pollution in the estuary, thus leading to this phenomenon. Interestingly, although the amount of river pollutants in August was greater than that in May 2017 (gray columns), Figure 4b still shows excellent water quality in the estuary of the south ZCS. In the summer, the ZFCC and Taiwan Warm Current (TWC) transport surface water to the northeast and take pollutants away with them, which may account for the high water quality in the south ZCS and the poor water quality in the north ZCS.[34] Furthermore, phytoplankton is more capable of absorbing nutrients in the summer, which may also mitigate the eutrophication in the south ZCS.[54]

In summary, it can be deduced that the poor water quality in the south ZCS in May 2017 resulted from the enhancement of riverine nutrients, and controlling the riverine pollution is an effective strategy for water quality improvement. Therefore, our method can reveal phenomena in the coastal water environment and provide decision support for water environmental management.

3.3.3. Water Quality Distribution in North ZCS. It is evident from the Figures 4 and 5 that the water quality around the Zhoushan Islands presents quite different spatial distributions, which might result from the different effects of the Yangtze River Diluted Water (YRDW). Previous theories demonstrated that the YRDW flows southward along the coast and directly affects the ZCS when the river velocity of the Yangtze River estuary is less than 3.8×10^4 m/s.[39] Otherwise, the main component of the YRDW changes direction and flows toward the northeast.[39] Therefore, we estimated the river velocity in the Yangtze River estuary and found that the velocities were 3.24×10^4, 3.40×10^4, and 3.09×10^4 m/s in May and August of 2015 and May 2017, respectively, and 5.01×10^4 m/s in May 2016 (Table S8). Accordingly, the YRDW changed direction in May 2016 and did not turn in the other three months. This is consistent with the inference that the water quality in Figure 4c presents much better conditions in comparison to those for the other three months because the YRDW probably shifted toward the northeast in May 2016.

Notably, Figure 4b shows a large area of level 15 water distributed around the Zhoushan Islands in August 2015. This is consistent with the reports from the OMFC that two algal blooms occurred near the Zhoushan Islands at the end of July 2015, indicating that the favorable temperature and rich nutrients in the summer increased algae breeding and polluted surface water around the Zhoushan Islands. In addition, as previously mentioned, the ZFCC and TWC might bring surface nutrients toward the center of the ZCS, which can also promote the accumulation of the algae.[34,54]

Consequently, the WQC maps reveal the water quality deterioration in the north ZCS that resulted from the effect of YRDW, and this effect probably expanded to the southern parts of the Zhoushan Islands. Moreover, climate changes such as other ocean currents, monsoon, and temperature variations also support the distribution shown in the maps.

3.4. Suggestions for Water Quality Management. This paper presents a novel water quality assessment method to facilitate the characterization of the spatiotemporal patterns

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of coastal water quality and the detection of abnormal regions. It also assesses the effects of terrestrial discharges and seasonal changes on the water environment in the ZCS. Further, the generated WQC maps can help provide targeted regulatory interventions with accurate spatial locations for the water quality management in the ZCS. For instance, environmental managers should pay attention to the management of tourist areas in inner Xiangshan Bay and industrial water control in Hangzhou Bay (especially in the region at 121.019°N, 30.297°E and 121.638°N, 30.153°E) and attempt to increase environmental awareness among enterprises and the public. In the north ZCS, the adverse effect of the Yangtze River is so strong that we suggest more sampling sites be added at the northernmost point of the ZCS (122.086°N, 30.778°E), and water flow should also be measured to permit investigation of the direction of YRDW. Due to the complex influences of terrestrial discharges and ocean currents, chlorophyll and bacterium should be monitored around the Zhoushan Islands to prevent their harmful effects on aquaculture and fisheries. To further improve the water quality in the estuaries of the south ZCS, pollutant inputs along the rivers should be strictly controlled and discharges from ships should be closely monitored.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.0c05928.

Detailed text information and figures and tables as described in the text (PDF)

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Notes

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